

AI/MCP-Driven Product Development System - Architecture Summary

This document outlines a system that automates the full product development lifecycle using AI and MCP integrations. It connects JIRA-based requirements to Figma wireframes, generates React/Tailwind UI code, creates Gherkin test cases, converts them into Playwright scripts, and enables CI/CD deployment.

C4 Architecture Overview

Level 1 - System Context:

- Product Owner interacts with AI/MCP Dev System.
- The system integrates with JIRA (for requirements), Figma (for design), and LLMs (for code/test generation).

Level 2 - Containers:

- Integration Layer, Requirements Parser, Prompt Engine
- Wireframe Generator, UI Code Generator, Gherkin Generator
- Playwright Test Runner, CI/CD Pipeline

Level 3 - Components:

- Prompt Engineering Module
- LLM Integration (Groq/OpenAI)
- UI Code Synthesizer (React + Tailwind)
- Gherkin Scenario Generator
- Playwright Script Generator

Mermaid Diagram (C4 Level 3)

```
graph TD
    subgraph Code_And_Test_Generator
        Prompt[Prompt Engineering Module]
        LLM[LLM Integration - Groq or OpenAI]
        CodeGen[UI Code Synthesizer - React and Tailwind]
        Gherkin[Gherkin Scenario Generator]
        Playwright[Playwright Script Generator]
```

end

Prompt --> LLM

LLM --> CodeGen

LLM --> Gherkin

Gherkin --> Playwright

Figma Plugin Feedback Flow

1. PO selects a component on the AI-generated wireframe.
2. Plugin presents options: Accept / Reject / Comment.
3. PO submits feedback tied to Figma node IDs.
4. JSON feedback is sent back to MCP backend to update layout or trigger code generation.

Prompt Engineering Example

Prompt:

Design a login screen with:

- Email (required), Password (required)
- Login button, Forgot password link

Generate layout using React + Tailwind CSS.